MP Modelling

Montag, 12. Oktober 2026 23:01

Q1.1 Kinematics (1 pt.)

Model the kinematics of the system. At the end you should have an expression of the x, y coordinates and velocities of the print head based on the motor angles θ_l and θ_r. We expect formulations for the variables:

$$x(\theta_l, \theta_r),\ y(\theta_l, \theta_r) \tag{1}$$

$$\dot{x}(\dot{\theta}_l, \dot{\theta}_r),\ \dot{y}(\dot{\theta}_l, \dot{\theta}_r). \tag{2}$$

Rotation at motor into translation

$s_l = r_m \theta_l \qquad s_r = r_m \theta_r$

If the rope moves it will result in a linear combination of x and y movement

left side will result in x and y positive

$s_l = x + y$

right side will result in x positive and y negative

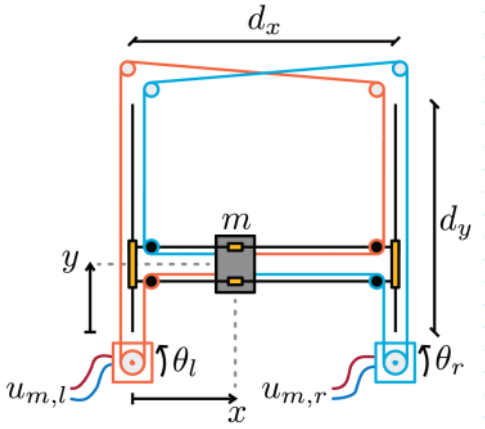
$s_r = x - y$

Add s_l and s_r for the expression for x and subtract s_r from s_l for y

$s_l + s_r = 2x \Rightarrow x = \frac{1}{2}(s_l + s_r) \qquad s_l - s_r = 2y \Rightarrow y = \frac{1}{2}(s_l - s_r)$

$\Rightarrow x(\theta_l, \theta_r) = \frac{r_m}{2}(\theta_l + \theta_r) \qquad y(\theta_l, \theta_r) = \frac{r_m}{2}(\theta_l - \theta_r)$

$\dot{x}(\dot{\theta}_l, \dot{\theta}_r) = \frac{r_m}{2}(\dot{\theta}_l + \dot{\theta}_r) \qquad \dot{y}(\dot{\theta}_l, \dot{\theta}_r) = \frac{r_m}{2}(\dot{\theta}_l - \dot{\theta}_r)$



Name	Symbol
Motor angle left	θ_l
Motor voltage left	$u_{m,l}$
Motor current left	$i_{m,l}$
Motor radius	r_m
Motor induction	L_m
Motor resistor	R_m
Motor inertia	J_m
Motor constant	K_m
Tool head position	x, y
Tool head mass	m
Nozzle voltage	u_n
Nozzle current	i_n
Nozzle capacitor	C_n
Nozzle thermal capacity	$C_{t,n}$
Nozzle thermal resistance	$R_{t,n}$
Nozzle resistor	R_n
Nozzle temperature	T_n
Environmental temperature	T_e
Bearing friction rolling	μ_r
Bearing friction linear x-direction	$\mu_{l,x}$
Bearing friction linear y-direction	$\mu_{l,y}$
Print bed temperature	T_b
Print area dimension x	d_x
Print area dimension y	d_y

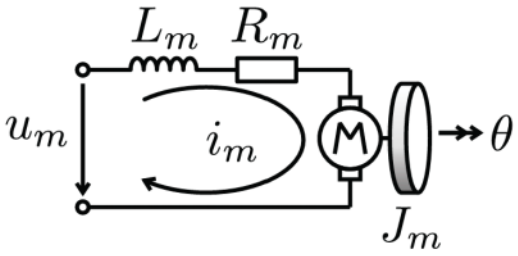
Q1.2 Dynamics (2 pts.)

Model the dynamics of the system. At the end of this part you should have a state-space description of the system with the voltages of the motors as inputs of the form

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}) \tag{3}$$

with $\boldsymbol{x} = [\theta_l, \theta_r, \cdots]^\top$ and $\boldsymbol{u} = [u_l, u_r]^\top$.

An important caveat! Your colleague ordered the wrong motors, so instead of the typical stepper motors, you only have voltage-controlled DC motors at your disposal. As a result, you have to consider the following circuit to model the motor:



Mechanical kinetic energy

$$K_{mech} = \frac{1}{2} m \left\| \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} \right\|^2 + \frac{1}{2} J_m \dot{\theta}_l^2 + \frac{1}{2} J_m \dot{\theta}_r^2 = \frac{1}{2} m \frac{r_m^2}{4} \left(\dot{\theta}_l^2 + 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2 + \dot{\theta}_l^2 - 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2 \right) + \frac{1}{2} J_m \left(\dot{\theta}_l^2 + \dot{\theta}_r^2 \right) = \frac{1}{2} \left(\dot{\theta}_l^2 + \dot{\theta}_r^2 \right) \left(\frac{m r_m^2}{2} + J_m \right)$$

Electrical „kinetic" energy (terms for electromagnetic coupling, don't forget citing)

$$K_{elec} = \frac{1}{2} L_m \dot{q}_{m,l}^2 + \frac{1}{2} L_m \dot{q}_{m,r}^2 + K_m \dot{q}_{m,l} \dot{\theta}_l + K_m \dot{q}_{m,r} \dot{\theta}_r = \frac{1}{2} L_m \left(\dot{q}_{m,l}^2 + \dot{q}_{m,r}^2 \right) + K_m \left(\dot{q}_{m,l} \dot{\theta}_l + \dot{q}_{m,r} \dot{\theta}_r \right)$$

→ with q being charge and therefore $\dot{q}$ equals the current

Mech. dissipative energy

$$D_{mech} = \frac{1}{2} \mu_{l,x} \dot{x}^2 + \frac{1}{2} \mu_{l,y} \dot{y}^2 + \frac{1}{2} \mu_r \dot{\theta}_l^2 + \frac{1}{2} \mu_r \dot{\theta}_r^2 = \frac{1}{2} \frac{\mu_{l,x} r_m^2}{4} \left(\dot{\theta}_l^2 + 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2 \right) + \frac{1}{2} \frac{\mu_{l,y} r_m^2}{4} \left(\dot{\theta}_l^2 - 2\dot{\theta}_l\dot{\theta}_r + \dot{\theta}_r^2 \right) + \frac{1}{2} \mu_r \left(\dot{\theta}_l^2 + \dot{\theta}_r^2 \right)$$

Elect. dissipative energy

$$D_{elec} = \frac{1}{2} R_m \dot{q}_{m,l}^2 + \frac{1}{2} R_m \dot{q}_{m,r}^2 = \frac{1}{2} R_m \left(\dot{q}_{m,l}^2 + \dot{q}_{m,r}^2 \right)$$

External forces

for $\dot{q}_{m,l}$: $u_{m,l}$

for $\dot{q}_{m,r}$: $u_{m,r}$

Lagrange equation

$$L = K - P = K_{mech} + K_{elec} = \frac{1}{2} \left(\dot{\vartheta}_l^2 + \dot{\vartheta}_r^2 \right) \left(\frac{m r_m^2}{2} + J_m \right) + \frac{1}{2} L_m \left(\dot{q}_{m,l}^2 + \dot{q}_{m,r}^2 \right) + K_m \left(\dot{q}_{m,l} \dot{\vartheta}_l + \dot{q}_{m,r} \dot{\vartheta}_r \right)$$

Total dissipative energy

$$D = D_{mech} + D_{elec} = \frac{1}{2} \frac{\mu_{l,r} r_m^2}{4} \left(\dot{\vartheta}_l^2 + 2\dot{\vartheta}_l \dot{\vartheta}_r + \dot{\vartheta}_r^2 \right) + \frac{1}{2} \frac{\mu_{l,y} r_m^2}{4} \left(\dot{\vartheta}_l^2 - 2\dot{\vartheta}_l \dot{\vartheta}_r + \dot{\vartheta}_r^2 \right) + \frac{1}{2} \mu_r \left(\dot{\vartheta}_l^2 + \dot{\vartheta}_r^2 \right) + \frac{1}{2} R_m \left(\dot{q}_{m,l}^2 + \dot{q}_{m,r}^2 \right)$$

to get $\frac{d}{dt} \frac{\partial L}{\partial \dot{v}_i} - \frac{\partial L}{\partial v_i} + \frac{\partial D}{\partial \dot{v}_i} = Q_i$, do partial derivatives with $v = [\vartheta_l, \vartheta_r, q_{m,l}, q_{m,r}]$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\vartheta}_l} = \frac{d}{dt} \left(\left(\frac{m r_m^2}{2} + J_m \right) \dot{\vartheta}_l \right) = \left(\frac{m r_m^2}{2} + J_m \right) \ddot{\vartheta}_l \qquad \frac{\partial L}{\partial \vartheta_l} = K_m \dot{q}_{m,l} \qquad \frac{\partial D}{\partial \dot{\vartheta}_l} = \frac{\mu_{l,r} r_m^2}{4} \dot{\vartheta}_l + \frac{\mu_{l,r} r_m^2}{4} \dot{\vartheta}_r + \frac{\mu_{l,y} r_m^2}{4} \dot{\vartheta}_l - \frac{\mu_{l,y} r_m^2}{4} \dot{\vartheta}_r + \mu_r \dot{\vartheta}_l$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\vartheta}_r} = \left(\frac{m r_m^2}{2} + J_m \right) \ddot{\vartheta}_r \qquad \frac{\partial L}{\partial \vartheta_r} = K_m \dot{q}_{m,r} \qquad \frac{\partial D}{\partial \dot{\vartheta}_r} = \frac{\mu_{l,r} r_m^2}{4} \dot{\vartheta}_l + \frac{\mu_{l,r} r_m^2}{4} \dot{\vartheta}_r - \frac{\mu_{l,y} r_m^2}{4} \dot{\vartheta}_l + \frac{\mu_{l,y} r_m^2}{4} \dot{\vartheta}_r + \mu_r \dot{\vartheta}_r$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_{m,l}} = \frac{d}{dt} \left(L_m \dot{q}_{m,l} + K_m \vartheta_l \right) = L_m \ddot{q}_{m,l} + K_m \dot{\vartheta}_l \qquad \frac{\partial L}{\partial q_{m,l}} = 0 \qquad \frac{\partial D}{\partial \dot{q}_{m,l}} = R_m \dot{q}_{m,l}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_{m,r}} = L_m \ddot{q}_{m,r} + K_m \dot{\vartheta}_r \qquad \frac{\partial L}{\partial q_{m,r}} = 0 \qquad \frac{\partial D}{\partial \dot{q}_{m,r}} = R_m \dot{q}_{m,r}$$

$$\vartheta_l : \quad \left(\frac{m r_m^2}{2} + J_m \right) \ddot{\vartheta}_l - K_m \dot{q}_{m,l} + \left(\frac{\mu_{l,r} r_m^2}{4} + \frac{\mu_{l,y} r_m^2}{4} + \mu_r \right) \dot{\vartheta}_l + \left(\frac{\mu_{l,r} r_m^2}{4} - \frac{\mu_{l,y} r_m^2}{4} \right) \dot{\vartheta}_r = 0$$

$$\vartheta_r : \quad \left(\frac{m r_m^2}{2} + J_m \right) \ddot{\vartheta}_r - K_m \dot{q}_{m,r} + \left(\frac{\mu_{l,r} r_m^2}{4} - \frac{\mu_{l,y} r_m^2}{4} \right) \dot{\vartheta}_l + \left(\frac{\mu_{l,r} r_m^2}{4} + \frac{\mu_{l,y} r_m^2}{4} + \mu_r \right) \dot{\vartheta}_r = 0$$

$$q_{m,l} : \quad L_m \dot{q}_{m,l} + K_m \dot{\vartheta}_l + R_m \dot{q}_{m,l} = u_{m,l}$$

$$q_{m,r} : \quad L_m \ddot{q}_{m,r} + K_m \dot{\vartheta}_r + R_m \dot{q}_{m,r} = u_{m,r}$$

$$\vec{x} = \left[\vartheta_l, \vartheta_r, \dot{\vartheta}_l, \dot{\vartheta}_r, \underbrace{i_{m,l}}_{\dot{q}_{m,l}}, \underbrace{i_{m,r}}_{\dot{q}_{m,r}} \right]^T \qquad \vec{u} = \left[u_{m,l}, u_{m,r} \right]^T$$

$$\dot{\vec{x}} = \begin{bmatrix} \dot{\vartheta}_l \\ \dot{\vartheta}_r \\ \frac{1}{\frac{m r_m^2}{2} + J_m} \left(K_m i_{m,l} - \left(\frac{\mu_{l,r} r_m^2}{4} + \frac{\mu_{l,y} r_m^2}{4} + \mu_r \right) \dot{\vartheta}_l - \left(\frac{\mu_{l,r} r_m^2}{4} - \frac{\mu_{l,y} r_m^2}{4} \right) \dot{\vartheta}_r \right) \\ \frac{1}{\frac{m r_m^2}{2} + J_m} \left(K_m i_{m,r} - \left(\frac{\mu_{l,r} r_m^2}{4} - \frac{\mu_{l,y} r_m^2}{4} \right) \dot{\vartheta}_l - \left(\frac{\mu_{l,r} r_m^2}{4} + \frac{\mu_{l,y} r_m^2}{4} + \mu_r \right) \dot{\vartheta}_r \right) \\ \frac{1}{L_m} \left(-K_m \dot{\vartheta}_l - R_m i_{m,l} + u_{m,l} \right) \\ \frac{1}{L_m} \left(-K_m \dot{\vartheta}_r - R_m i_{m,r} + u_{m,r} \right) \end{bmatrix}$$

$$J = \frac{m r_m^2}{2} + J_m \qquad \mu_\Delta = \frac{\mu_{l,r} r_m^2}{4} - \frac{\mu_{l,y} r_m^2}{4} \qquad \mu = \frac{\mu_{l,r} r_m^2}{4} + \frac{\mu_{l,y} r_m^2}{4} + \mu_r$$

$$\dot{\vec{x}} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{\mu}{J} & -\frac{\mu_\Delta}{J} & \frac{K_m}{J} & 0 \\ 0 & 0 & -\frac{\mu_\Delta}{J} & -\frac{\mu}{J} & 0 & \frac{K_m}{J} \\ 0 & 0 & -\frac{K_m}{L_m} & 0 & -\frac{R_m}{L_m} & 0 \\ 0 & 0 & 0 & -\frac{K_m}{L_m} & 0 & -\frac{R_m}{L_m} \end{bmatrix} \vec{x} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_m} & 0 \\ 0 & \frac{1}{L_m} \end{bmatrix} \vec{u}$$

Q1.3 *Decouple the x and y dynamics (1 pt.)*

Investigate how you can transform the state-space model to decouple the x and y motions. You should arrive at two decoupled state-space descriptions:

$$\dot{\boldsymbol{x}}_x = f_x(\boldsymbol{x}_x, \boldsymbol{u}_x)\,, \qquad \dot{\boldsymbol{x}}_y = f_y(\boldsymbol{x}_y, \boldsymbol{u}_y)\,, \tag{4}$$

with $\boldsymbol{x}_x = [x, \cdots]^\top$, $\boldsymbol{x}_y = [y, \cdots]^\top$, and $\boldsymbol{u}_x$, $\boldsymbol{u}_y$ being functions of u_l and u_r.

Hint: *Review your solution from Q1.1.*

$x = \frac{r_m}{2}(\vartheta_l + \vartheta_r) \qquad y = \frac{r_m}{2}(\vartheta_l - \vartheta_r)$

$\ddot{x} = \frac{r_m}{2}\frac{1}{J}\left(-\mu\dot{\vartheta}_l - \mu_a\dot{\vartheta}_r + K_m i_{m,l} - \mu_a\dot{\vartheta}_l - \mu\dot{\vartheta}_r + K_m i_{m,r}\right) = \frac{r_m}{2J}\left(-(\mu+\mu_a)\underbrace{(\dot{\vartheta}_l + \dot{\vartheta}_r)}_{=\frac{2}{r_m}\dot{x}} + K_m\underbrace{(i_{m,l} + i_{m,r})}_{=\frac{2}{r_m}i_x}\right) = -\frac{\mu+\mu_a}{J}\dot{x} + \frac{K_m}{J}i_x$

$\ddot{y} = \frac{r_m}{2}\frac{1}{J}\left(-\mu\dot{\vartheta}_l - \mu_a\dot{\vartheta}_r + K_m i_{m,l} + \mu_a\dot{\vartheta}_l + \mu\dot{\vartheta}_r - K_m i_{m,r}\right) = \frac{r_m}{2J}\left(-(\mu-\mu_a)\underbrace{(\dot{\vartheta}_l - \dot{\vartheta}_r)}_{=\frac{2}{r_m}\dot{y}} + K_m\underbrace{(i_{m,l} - i_{m,r})}_{=\frac{2}{r_m}i_y}\right) = -\frac{\mu-\mu_a}{J}\dot{y} + \frac{K_m}{J}i_y$

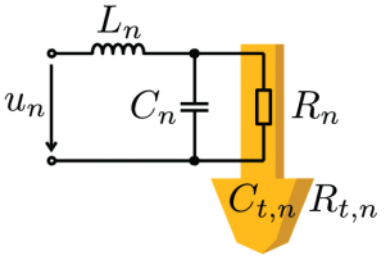
$\frac{di_x}{dt} = \frac{r_m}{2}\frac{1}{L_m}\left(-K_m\dot{\vartheta}_l - R_m i_{m,l} + u_{m,l} - K_m\dot{\vartheta}_r - R_m i_{m,r} + u_{m,r}\right) = \frac{r_m}{2}\frac{1}{L_m}\left(-K_m(\dot{\vartheta}_l + \dot{\vartheta}_r) - R_m(i_{m,l} + i_{m,r}) + u_{m,l} + u_{m,r}\right) = -\frac{K_m}{L_m}\dot{x} - \frac{R_m}{L_m}i_x + \frac{1}{L_m}u_x$

$\frac{di_y}{dt} = -\frac{K_m}{L_m}\dot{y} - \frac{R_m}{L_m}i_y + \frac{1}{L_m}u_y$

$$\Rightarrow \quad \dot{\vec{x}}_x = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{\mu+\mu_a}{J} & \frac{K_m}{J} \\ 0 & -\frac{K_m}{L_m} & -\frac{R_m}{L_m} \end{bmatrix}\begin{bmatrix} x \\ \dot{x} \\ i_x \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix} u_x \qquad \dot{\vec{x}}_y = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{\mu-\mu_a}{J} & \frac{K_m}{J} \\ 0 & -\frac{K_m}{L_m} & -\frac{R_m}{L_m} \end{bmatrix}\begin{bmatrix} y \\ \dot{y} \\ i_y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix} u_y$$

Q1.5 *Model the Nozzle (1 pt.)*

The nozzle is heated by a resistor and is subject to heat exchange with the environment and the filament. Input to the circuit is a controllable voltage. Model the governing differential equation(s). At the



end, you should have a state-space discription of the form

$$\dot{\boldsymbol{x}}_n = f_n(\boldsymbol{x}_n, u_n) \tag{5}$$

with $\boldsymbol{x} = [T_n, \cdots]^\top$. You may combine the thermal loss to the environment and the filament into one loss with respect to the environmental temperature T_e, which you can choose.

$u_n = L_n\frac{di_n}{dt} + u_c \quad \Rightarrow \quad \frac{di_n}{dt} = \frac{u_n - u_c}{L_n}$

$C_n\frac{du_c}{dt} = i_c \qquad \text{with } i_c = i_n - \frac{u_c}{R_n} \quad \Rightarrow \quad \frac{du_c}{dt} = \frac{i_n}{C_n} - \frac{u_c}{R_n C_n}$

$W = C_{t,n}\dot{T}_n \qquad P_{in} = \frac{u_c^2}{R_n} \qquad P_{out} = \frac{T_n - T_e}{R_{t,n}}$

$W = P_{in} - P_{out} = \frac{u_c^2}{R} - \frac{T_n - T_e}{\rho} = C_{t,n}\dot{T}_n \quad \Rightarrow \quad \dot{T}_n = \frac{u_c^2}{R_n C_{t,n}} - \frac{T_n - T_e}{R_{t,n} C_{t,n}}$

$$W = C_{t,n}\dot{T}_n \qquad P_{in} = \frac{u_c^2}{R_n} \qquad P_{out} = \frac{T_n - T_e}{R_{t,n}}$$

$$W = P_{in} - P_{out} = \frac{u_c^2}{R_n} - \frac{T_n - T_e}{R_{t,n}} = C_{t,n}\dot{T}_n \Rightarrow \dot{T}_n = \frac{u_c^2}{R_n C_{t,n}} - \frac{T_n - T_e}{R_{t,n} C_{t,n}}$$

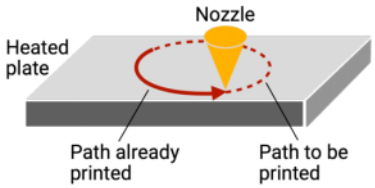
$$\vec{x}_n = [T_n, u_c, i_n]$$

$$\dot{\vec{x}}_n = \begin{bmatrix} \frac{u_c^2}{R_n C_{t,n}} - \frac{T_n - T_e}{R_{t,n} C_{t,n}} \\ \frac{i_n}{C_n} - \frac{u_c}{R_n C_n} \\ \frac{u_n - u_c}{L_n} \end{bmatrix}$$

Q1.6 Printing a circle of filament (PDE) (2 pts.)

When printing a 3D object layer-by-layer, care must be taken to manage the temperature of the solidifying filament already deposited, ensuring it is optimal before printing the next layer on top.

In the following figure, you see the printing of the first layer of a cylinder having an elliptical cross-section. The filament is deposited onto a heated base plate to ensure it does not cool too quickly.



Develop a 1D distributed-parameter model of the filament cooling down, and solve it numerically to show how long it takes for it to cool. This information is useful for planning the print of the next layer.

When modelling, you can assume for simplicity a square cross section of the printed filament. The lower side of it is in contact with the heated plate, while the other sides are exposed to the air at ambient temperature. The simulation should start as soon as the filament is being printed, which means your computational domain is growing over time. When the entire path has been printed, no more filament is deposited, but the simulation should continue until the temperature of the printed filament is nearly uniform.

